DHBW Ravensburg

Intelligence Engineering

Solving Real World Problems At Scale

Prof. Dr.-Ing. Mark Schutera | November 5, 2026 · *drifting stream*

LECTURE _

Orientation

what this course is, what it assumes, what you are going to know afterwards.

You have been given a lot of raw **data**,

and you want to engineer the shit out of it.

You have been given a business problem,

and you want to choose the right metrics and objectives to solve it.

Your initial models perform well in offline experiments,

and you want to deploy them at scale.

You want to harden your deployments

to quickly detect, debug, and address issues that arise in production.

There is a lot of manual work and handling involved so far,

and you want an operation vacation for your machine learning product or service.

You want to learn about a framework and foundation for building machine learning systems

that can be shared and reused across machine learning use cases and data products.

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Overview

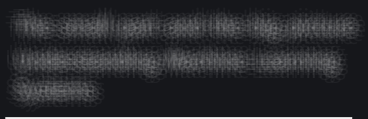
Machine Learning and Data Engineering

The lecture in one slide

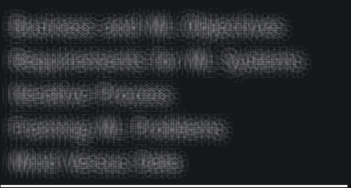
Lecture	Contents
I	Overview of Machine Learning Systems
II	Introduction to Machine Learning Systems Design
III	Data Engineering Fundamentals
IV	Training Data
V	Feature Engineering
VI	Model Development and Offline Evaluation
VII	Model Deployment and Prediction Service
VIII	Data Distribution Shifts and Monitoring
IX	Continual Learning and Test in Production
X	Infrastructure and Tooling for MLOps
XI	The Human Side of Machine Learning

The lecture in one slide

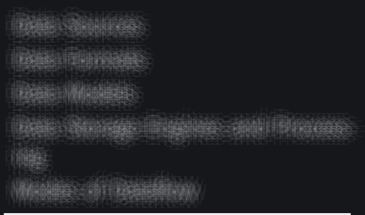
I Overview of Machine Learning Systems



II Introduction to Machine Learning Systems Design



III Data Engineering Fundamentals



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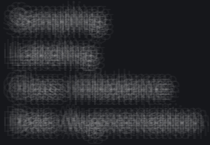
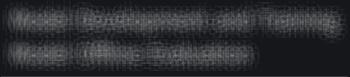
IV Training Data



V Feature Engineering

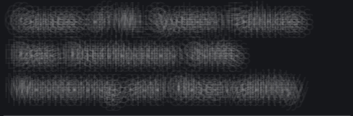
VI Model Development and Offline Evaluation



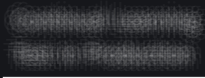
The lecture in one slide

VII Model Deployment and Prediction Service

VIII Data Distribution Shifts and Monitoring



IX Continual Learning and Test in Production



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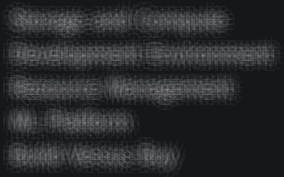
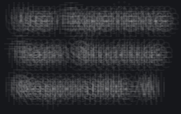
X Infrastructure and Tooling for MLOps



XI The Human Side of Machine Learning



Course Work

The Hive is Real. Dataset on Hugging Face.

Presentation (15 min) + Q&A (open)

You get recordings of a beehive entrance and contribute data engineering, an annotation policy, a prediction service or specialized tooling.

Open whiteboard, three slides or one page only. Be ready when the class starts.

The bee

Course Work

What I am looking for:

Introduction by Motivation and Literature

Method by Explanation and/or Demonstration

Results show what was achieved and how well

Discussion on findings, limitations and what's next

Smart and Aesthetic Thoughts, Designs, Ideas and Questions.

The bee

Extra Mile

Special projects

If you are up for the extra mile, let me know. I have special projects that I can assign to you.

Wrapped in the same presentation and Q&A format. Yes, it will be more challenging and take more time and effort.

A classic: lose-lose in the short term, yet **win-win in the long term**.

A large banana

Brainwriting, 90 minutes

1. **5 min** Form groups of **4 to 5** students.
2. **10 min** Research: Google, Agent, you name it.
3. **10 min** Each student writes four ideas.
4. **5 min** Research.
5. **10 min** Stacks pass to the next table, each adds two.
6. **5 min** Research.
7. **10 min** Stacks pass again, each adds the last two.
8. **15 min** Cluster and clean the ideas per group.
9. **15 min** Cluster across teams.
10. **5 min** Walk the idea deck: everyone looks at all clusters.

The bee

It is your obligation to

get yourself a presentation slot and brainstorm a project idea to work on.

LECTURE I

Overview of a Machine Learning System

This lecture in one slide

I Overview of Machine Learning Systems

The small part and the big picture

Understanding Machine Learning Systems

A **model** is only a small part from the system around it, and some might argue there are a lot of moments when not to reach for machine learning at all.

1

The small part and the big picture

Don’t confuse the model with the machine learning system.

Enabling the delivery of the logic

The **model** is the small part of a machine learning system. With foundation models and AI engineering, that part got smaller still. The essential work of a machine learning system remains and stacks around business understanding, infrastructure, design pattern and data engineering.

Before learning about solving a problem with machine learning, let's first assess when it is actually a good idea to do so.

(1) Learn

The problem you are solving, requires your system to have the capability to **learn**. Learning can look different depending on the problem and its context and boundary conditions. You already know about supervised learning, unsupervised learning and reinforcement learning paradigms.

Machine learning is an approach to (1) learn (2) complex patterns from (3) existing data and use these patterns to make (4) predictions on (5) unseen data.

(2) Pattern

Learning requires the existence of a **pattern**, as opposed to noise or randomness. It is not trivial nor obvious whether there is a pattern in the data, and whether it is learnable. Complex pattern is more difficult to embed or state explicitly, this is where machine learning has an edge over classical approaches.

Machine learning is an approach to (1) learn (2) complex patterns from (3) existing data and use these patterns to make (4) predictions on (5) unseen data.

(3) Data

Data is available, accessible or acquirable. The data is of **sufficient** quality and quantity to support learning, (and to validate for that matter). If your approach does not require data, it is because you build on embeddings or foundation models, which themselves are built on data in the first place, or you need to rely on data acquisition and curation as part of your system.

Machine learning is an approach to (1) learn (2) complex patterns from (3) existing data and use these patterns to make (4) predictions on (5) unseen data.

(4) Predictions

Machine learning is about making predictions, **always**. Predictions can be of different types, such as classification, regression, ranking, clustering, generation and on and on. By reframing problems in terms of predictions, we can leverage machine learning to solve them. Predictions are approximations by nature and therefore come with uncertainty, which needs to be accounted for.

Machine learning is an approach to (1) learn (2) complex patterns from (3) existing data and use these patterns to make (4) predictions on (5) unseen data.

(5) Unseen Data

In a closed system or domain, the data is finite and the system can be fully specified by optimization. However, often the domain is open and infinite, and the system needs to be able to **generalize** to unseen data. Of course, the unseen data is not completely unknown, it is drawn from the same distribution (i.i.d.) as the training data and is thus able to generalize to the unseen data inside a specific domain or distribution.

Machine learning is an approach to (1) learn (2) complex patterns from (3) existing data and use these patterns to make (4) predictions on (5) unseen data.

Catalysts

A catalyst is a substance that increases the rate of reaction without undergoing change itself.

The task is repetitive.

In a sense that we have a lot of data to learn a pattern from.

It scales.

The need to scale a specific prediction capability, not only a few times.

The context changes.

Meaning that hard coded or other solutions will be brittle and break.

Being wrong is not too bad.

Approximations on unseen data. Failures need to be manageable.

When Not to Use Machine Learning

It is unethical.

Well, that might also be a general rule. But doing stuff at scale, without human oversight, has its implications.*

There's a simple solution.

A simple solution, heuristic, or baseline should always be the first step regardless.

It's not economically viable.

The value of the prediction, needs to be greater than the cost of the system, including development, deployment and maintenance.

* More? Enroll to my Philosophy of AI course.

Describe two use-cases and the rationale for whether or not to use machine learning.

One Machine Learning Use-Case and One which is not.

The field is moving, 2026

Adoption is near universal, value capture is not.

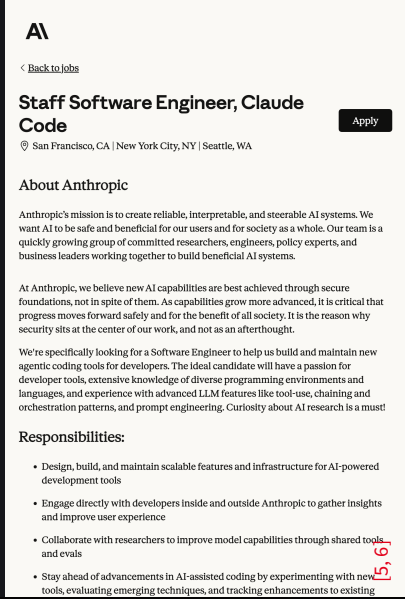
- **Adoption** 88% of organizations use AI in at least one function, 72% use gen AI.
- **Reach** Gen AI reached 53% of the population in three years, faster than the PC or the internet.
- **Productivity** 80% of users report personal gains.
- **Slop** 40% of desk workers say they received AI workslop in the past month. Each case costs almost two hours to fix, about $186 per employee per month.
- **Value** Only **37%** of organizations attribute any EBIT impact to AI, flat since 2025, and 6% qualify as high performers.

[2–4]

“… in twelve months, we may be in a world where AI is writing essentially all of the code.”

Dario Amodei, March 2025. Nineteen months later, Anthropic lists 94 open software engineer roles.

Anthropic hiring, October 2026



‹ Back to jobs

Staff Software Engineer, Claude Code

Apply

San Francisco, CA | New York City, NY | Seattle, WA

About Anthropic

Anthropic's mission is to create reliable, interpretable, and steerable AI systems. We want AI to be safe and beneficial for our users and for society as a whole. Our team is a quickly growing group of committed researchers, engineers, policy experts, and business leaders working together to build beneficial AI systems.

At Anthropic, we believe new AI capabilities are best achieved through secure foundations, not in spite of them. As capabilities grow more advanced, it is critical that progress moves forward safely and for the benefit of all society. It is the reason why security sits at the center of our work, and not as an afterthought.

We're specifically looking for a Software Engineer to help us build and maintain new agentic coding tools for developers. The ideal candidate will have a passion for developer tools, extensive knowledge of diverse programming environments and languages, and experience with advanced LLM features like tool-use, chaining and orchestration patterns, and prompt engineering. Curiosity about AI research is a must!

Responsibilities:

- Design, build, and maintain scalable features and infrastructure for AI-powered development tools
- Engage directly with developers inside and outside Anthropic to gather insights and improve user experience
- Collaborate with researchers to improve model capabilities through shared tools and evals
- Stay ahead of advancements in AI-assisted coding by experimenting with new tools, evaluating emerging techniques, and tracking enhancements to existing

[5, 6]

2

Understanding Machine Learning Systems

Research and Production are not the same.

Horizontal versus Vertical

Nothing motivates system thinking more than the need to go horizontal. Academia is vertical, with a narrow focus on a single problem, where drilling deep, fast is rewarded and focus allows to teach a wide range of concepts.

	Research	Production
Objectives	State of the art, highly engineered	Stakeholder outcome driven
Compute	Training and high throughput	Fast inference with low latency
Data	Often static benchmark oriented	Shifting domains and structures
Usability metrics	Usually not considered	UI, Fairness, Robust, Interpretability

Stakeholders and Objectives

In research the objective is one number on one benchmark. Ship the same model as a product and five people want five different things from it: **conflicting requirements** are the norm, and not all of them are must haves.

ML engineers | Platform team | Product team | Sales team | Manager

The bee hive

Different stakeholders have different objectives and think in different solution spaces.

Define the next step for the bee hive and line out how to reach that.

Five people, five solution spaces

Each of them is right. The **objective** is what they are measured on, the solution space is where they will look for it.

	Objective	Solution space
ML engineers	better bee counts	more labelled frames, a bigger model
Platform team	nothing pages at 3am	freeze the model, build the platform
Product team	the hive view feels live	a latency budget of 100 ms
Sales team	revenue per hive	an alert for every hive, sold as premium
Manager	margin	fewer models, and maybe fewer engineers

Computational priorities

Research optimises **throughput**, production optimises **latency**.

Processing	Latency	Throughput
One query at a time	10 ms	100 queries/s
Batches of 10	20 ms	500 queries/s
Batches of 100	50 ms	2,000 queries/s

Every row is the same model on the same hardware. Only the batch changed.

Batching waits for the queue to fill, so each single request waits longer while the machine as a whole serves far more of them.

Batch size is not the only lever. Model size, sample size and how many models sit in one request all move the numbers.

Latency is a distribution

Ten requests, latency in ms:

10.0, 9.7, 10.4, **90.0**, 9.9, 10.0, 11.0, 9.5, 10.2, 10.0

The mean would be 18.1 ms. Nine of the ten requests were served in about 10 ms. Mean only holds at larger scales.

2017, Akamai: A 100 ms delay can cost 7% of conversions.

2019, Booking.com: about 30% more latency cost about 0.5% in conversion.

2016, Google: Over half of mobile users leave a page that takes more than three seconds.

[6-7]

Percentiles, not averages

The same ten requests, sorted. A percentile is a kind of promise: q percent of requests finish in p_q or less.

9.5, 9.7, 9.9, 10.0, 10.0 (p_{50}), 10.0, 10.2, 10.4, 11.0 (p_{90}), **90.0** (p_{99})

$$p_q = \min\left\{ t : \frac{\#\,\text{requests} \le t}{n} \ge \frac{q}{100} \right\}$$

The mean of 18.1 ms moves with the outlier, the percentile makes the outlier visible.

With ten requests p_{99} is just the maximum. It takes 100 requests before p_{99} and the maximum can differ. Precision needs to be bought with data.

Quality gates count in nines: Three nines is $p_{99.9}$ (1 in 1000 requests), five nines $p_{99.999}$ (1 in 100000).

[10]

Your toxic relationship with mean accuracy

Latency is one metric. It might not be yours to measure. Latency is a distribution. As is any metric. So if you are only reporting mean accuracy, you are **underdelivering** already.

Start with the feature the solution has to deliver on. Pick the metrics that speak to them. Then curate the data that allows to measure them. Only then build the methods that move those metrics.

[11]

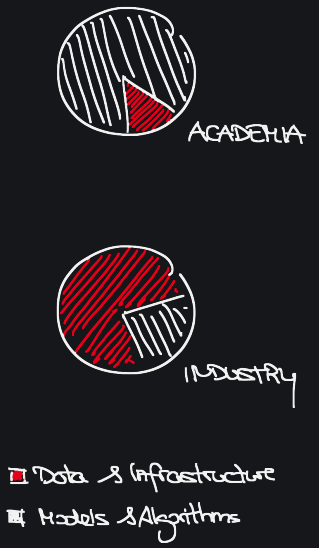
Data in research

A research dataset is a teaching artefact. Somebody already did the work, for something very specific.

- Clean and well formatted
- Static, so the community can benchmark against it
- Quirks of the dataset are known and documented
- Open source scripts to feed your models
- Historical, already stored somewhere
- Reported results usually overfit

Titanic [12]: 891 rows, 12 columns.

Karpathy: At Tesla you lose sleep over data



[13]

Leaderboards and Benchmarks

Well curated data is expensive and cumbersome. Benchmarks allow comparability, come at scale, and usually link proper metrics and state of the art methods. And they can be adjusted.

- System This scores a model, and a model is not a system.
- Metric It measures not necessarily what you care about.
- Overfit Every run against a benchmark stales it, fast.
- Chance Seeds move CIFAR 10 by 1.82 points.
- Problem Benchmarks are a great proxy, but usually not your problem exactly.

GLUE 2018. Trust me bro, nine tasks

GLUE: A Multi-Task Benchmark and Analysis Platform for Natural Language Understanding

Alex Wang[1], **Amanpreet Singh**[1], **Julian Michael**[2], **Felix Hill**[3], **Omer Levy**[2], **and Samuel R. Bowman**[1]

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Human ability to understand language is *general, flexible*, and *robust*. In contrast, most NLU models above the word level are designed for a specific task and struggle with out-of-domain data. If we aspire to develop models with understanding beyond the detection of superficial correspondences between inputs and outputs, then it is critical to develop a unified model that can execute a range of linguistic tasks across different domains.

To facilitate research in this direction, we present the General Language Understanding Evaluation (GLUE, gluebenchmark.com): a benchmark of nine diverse NLU tasks, an auxiliary dataset for probing models for understanding of specific linguistic phenomena, and an online platform for evaluating and comparing models. For some benchmark tasks, training data is plentiful, but for others it is limited or does not match the genre of the test set. GLUE thus favors models that can represent linguistic knowledge in a way that facilitates sample-efficient learning and effective knowledge-transfer across tasks. While none of the datasets in GLUE were created from scratch for the benchmark, four of them feature privately-held test data, which is used to ensure that the benchmark is used fairly.

We evaluate baselines that use ELMo (Peters et al., 2018), a powerful transfer learning technique, as well as state-of-the-art sentence representation models. The best models still achieve fairly low absolute scores. Analysis with our diagnostic dataset yields similarly weak performance over all phenomena tested, with some exceptions.

The GLUE benchmark GLUE consists of nine English sentence understanding tasks covering a broad range of domains, data quantities, and difficulties. As the goal of GLUE is to spur development of models that can quickly adapt to and learn to solve multiple tasks, we choose tasks with at least 2k examples, and favor data in slightly higher resource regimes. We also strongly favor tasks with multi-sentence contexts, to test models' ability to learn long-range dependencies. Following the genre of the GLUE tasks, test labels are hidden, and evaluation is mediated through a server. We describe the tasks briefly here, and in detail in Appendix A.

Corpus	|Train|	Task	Domain
Single-Sentence Tasks			
CoLA	8.5k	acceptability	misc.
SST-2	67k	sentiment	movie reviews
Similarity and Paraphrase Tasks			
MRPC	3.7k	paraphrase	news
STS-B	7k	textual sim.	misc.
QQP	364k	paraphrase	online QA
Inference Tasks			
MNLI	393k	NLI	misc.
QNLI	108k	QA/NLI	Wikipedia
RTE	2.5k	NLI	misc.
WNLI	634	coref./NLI	fiction books

Table 1: Task descriptions and statistics. **Bold** denotes tasks for which there is privately-held test data. All tasks are binary classification, except STS-B (regression) and MNLI (three classes).

The nine sentence understanding tasks, by type:

Single-Sentence Tasks

CoLA The Corpus of Linguistic Acceptability (Warstadt et al., 2018) consists of English acceptability judgments drawn from books and journal articles on linguistic theory. Each example is a sequence of words annotated with whether it is a grammatical English sentence. Following the original work, we use Matthews correlation coefficient (Matthews, 1975) as the evaluation metric, with thresholding for binary judgments.

SST-2 The Stanford Sentiment Treebank (Socher et al., 2013) consists of sentences from movie reviews and human annotations of their sentiment. The task is to predict the sentiment of a given sentence. We use the two-way (positive/negative) class split, with only sentence-level labels.

Similarity and Paraphrase Tasks

MRPC The Microsoft Research Paraphrase Corpus (Dolan and Brockett, 2005) is a corpus of sentence pairs automatically extracted from online news sources, with human annotations for whether the sentences in the pair are semantically equivalent. Because the classes are imbalanced (68% positive), we follow common practice and report both accuracy and F1 score.

STS-B The Semantic Textual Similarity Benchmark (Cer et al., 2017) is a collection of sentence pairs drawn from news headlines, video and image captions, and natural language inference data. Each pair is human-annotated with a similarity score from 1 to 5.

QQP The Quora Question Pairs dataset is a collection of question pairs from the community question-answering website Quora. The task is to determine whether a pair of questions are semantically equivalent. As in MRPC, the class distribution in QQP is unbalanced (63% negative), so we report both accuracy and F1 score.

Inference Tasks

MNLI The Multi-Genre Natural Language Inference Corpus (Williams et al., 2018) is a crowdsourced collection of sentence pairs with textual entailment annotations. Given a premise sentence and a hypothesis sentence, the task is to predict whether the premise entails the hypothesis (entailment), contradicts the hypothesis (contradiction), or neither (neutral). The premise sentences are gathered from ten different sources, including transcribed speech, fiction, and government reports. We use the standard test set, for which we obtain private labels from the author, and evaluate on both the matched (in-domain) and mismatched (cross-domain) sections. We also use and recommend the SNLI corpus (Bowman et al., 2015) as 550k examples of auxiliary training data.

QNLI The Question-answering NLI dataset is converted from the Stanford Question Answering Dataset (Rajpurkar et al., 2016). The task is to determine whether a context sentence contains the answer to a question sentence. We modify the original task to a binary classification task where the positive label is an accurate answer to the question, and the negative label is any context sentence that does not contain the correct answer. We construct positive examples by selecting each sentence that answers a question and pairing it with the question to create a dataset with balanced classes.

RTE The Recognizing Textual Entailment (RTE) datasets come from a series of annual textual entailment challenges. We combine the data from RTE1 (Dagan et al., 2006), RTE2 (Bar Haim et al., 2006), RTE3 (Giampiccolo et al., 2007), and RTE5 (Bentivogli et al., 2009). Examples are constructed based on news and Wikipedia text. We convert all datasets to a two-class split, where for three-class datasets we collapse neutral and contradiction into not entailment, for consistency.

WNLI The Winograd NLI dataset is adapted from the Winograd Schema Challenge (Levesque et al., 2012). The task is to predict if the sentence with a pronoun substituted is entailed by the original sentence. We use a small evaluation set consisting of new examples derived from fiction books that was shared privately by the authors of the original corpus. While the included training set is balanced between two classes, the test set is imbalanced between them (65% not entailment). Also, due to a data quirk, the development set is adversarial: hypotheses are sometimes shared between training and development examples, so if a model memorizes the training examples, it will predict the wrong label on corresponding development set example. As with QNLI, each example is evaluated separately, so there is not a systematic correspondence between a model's score on this task and its score on the unconverted original task. We call the converted dataset WNLI (Winograd NLI).

Evaluation For each task, we evaluate performance with the standard metric for that task. If a task has multiple metrics (e.g., accuracy and F1), we use the unweighted average of the metrics. We then compute a macro-average of the scores for all tasks to obtain a final score. We exclude WNLI from the average, due to the issues mentioned above.

Diagnostic Dataset Drawing inspiration from the FraCaS suite (Cooper et al., 1996) and other inference evaluation datasets, we include a small hand-crafted dataset for fine-grained analysis of system performance on a broad range of linguistic phenomena. This dataset is evaluated using the model that was trained for the MNLI task, and examples are scored using the MNLI scoring function, which is the trivalent classification accuracy. We call this dataset the GLUE Diagnostic Dataset. Since it is meant purely for evaluation, we only provide a test set. Examples are labeled with a three-way entailment label (entailment, neutral, or contradiction) and tagged with labels that indicate the linguistic phenomena that characterize each example. We describe the construction of this dataset in Section 4.

Platform We build GLUE around a platform hosted at gluebenchmark.com. The platform is model-agnostic and allows users to evaluate any system that can produce results on the GLUE tasks. Users can see a live leaderboard of scores on the GLUE tasks, and download the GLUE data with a single command. The platform also includes a manual error analysis page, which we use in Section 4.

Desiderata We design GLUE to favor models that share general linguistic knowledge across tasks. Specifically, we design GLUE to favor models that share statistical information in the form of parameters or learned representations. We do not explicitly favor models that share architectural components across tasks, but we do favor models that can train effectively in a multi-task setting. We also favor models that can learn to handle multiple tasks without forgetting how to handle earlier tasks. We do not favor models that use task-specific components, but we do favor models that can learn to handle multiple tasks without task-specific components. Though it is possible to train a model per task and evaluate the resulting set of models on this benchmark, we expect that inclusion of several data-scarce tasks will ultimately render this approach uncompetitive.

The nine tasks include two tasks with single-sentence inputs: Corpus of Linguistic Acceptability (CoLA; Warstadt et al. 2018) and Stanford Sentiment Treebank (SST-2; Socher et al. 2013). Three tasks involve detecting semantic similarity: Microsoft Research Paraphrase Corpus (MRPC; (Dolan and Brockett, 2005)), Quora Question Pairs[1] (QQP), and Semantic Textual Similarity Benchmark (STS-B; Cer et al. 2017). The remaining four tasks are formatted as natural language inference (NLI) tasks, such as the Multi-Genre NLI corpus (MNLI; Williams et al. 2018) and Recog-

Data in production

Nobody did the work. The data is a by-product of a business that was not run for your benefit. Usually you don't get away with building and curating your own dataset to gain back control and clear the fog.

- Messy, noisy, possibly unstructured
- Likely biased, and you need to find out how
- Labels sparse, imbalanced, or incorrect
- Changing requirements can mean relabeling
- Privacy and regulatory concerns
- Generated by users, systems, third parties

LLM as a judge (is the new ”I tested on the training data”)

An LLM judge grades your model with another model. Against its labels $\tilde{y}$ instead of the true y, your accuracy is off by as much as the judge is wrong.

Never for **validation**. For training data at most, and only while it adds no more $\tilde{y}$ than your ground truth already has.

- **Hallucination** grades claims it cannot check
- **Position bias** swap the answers, 65% of verdicts hold
- **Self-preference** rates its own, and its family’s, outputs higher

The judge meets the judged [17–19]

Perpetuating Bias

Machine learning algorithms do not predict the future, they encode the past, and then they act on it.

The invisible hand of performative prediction plays out t scale: With an increase in agentic systems the prediction is not merely a reading of the world, it is a push on it.

2014 to 2018, Amazon built a résumé ranker on ten years of its own hiring. It learned to penalise the word “women’s” and graduates of two women’s colleges. The project was scrapped.

2015, Google Photos labelled two Black users as gorillas. The fix shipped, and three years later, was still to delete the label from the vocabulary.

[20–23]

Interpretability

Research rewards performance, not interpretability. Industry asks for it, but is anyway slow to build it:

- **Users**, business leaders and end users alike, to see why a decision was made, so they can trust it and catch what is wrong with it.
- **Developers**, to debug, improve and align the model.

Most of us use a fridge without understanding it. Fewer feel that way about the thing deciding on their mortgage. Only 19% of large companies are working on explainability. Jev, 2026, is built only to decide: a typed answer with a probability, and no text to read an explanation from.

Press 2: Hand Claude recursive permission

[24]

Human on the loop

Putting a human on the loop is the standard answer to an uninterpretable or weak model. The evidence that it works is **thin**.

Supervising reliable automation degrades exactly the capability being supervised: Vigilance falls, response to the first failure is poor, and the takeover you designed for arrives too late.

Tempe, 2018: the system saw the pedestrian 5.6 s before impact and handed over at 1.2 s. The safety driver, eyes on her phone for 34% of the drive, looked up 1 s before. Test a human on the loop like any other design decision.

Tempe Police dashcam, March 2018

Why this is different from traditional software engineering

Traditional software engineering tests and versions code. A machine learning system has to test and version code, data, and everything the two produce together.

- How do you version a dataset, and diff two versions of one?
- How do you deploy into an environment that keeps moving?
- How do you monitor and debug a system with no stack trace?
- How do you avoid being harmful at scale and even push it?

Revisit the craft in **Full Stack Handwerkszeug**: Git, development and production, CI/CD.

Two books worth owning,

Software Engineering at Google [26]: engineering is programming integrated over time. Every question on the left is a question about time.

The Mythical Man-Month [27]: essential complexity against accidental. Data is essential complexity, and there is no silver bullet for it either.

A trained model is the small box in the middle.

You might argue that the engineering and development side of machine learning is more exciting.

Most jobs will be, and already are in productionizing machine learning systems or decision making.

LECTURE II

Introduction to Machine Learning Systems Design

This lecture in one slide

II Introduction to Machine Learning Systems Design

Business and ML Objectives

Requirements for ML Systems

Iterative Process

Framing ML Problems

Mind Versus Data

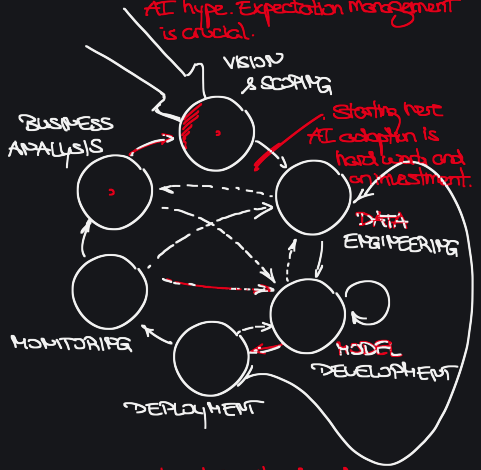
1

Business and ML Objectives

First Guidance on Developing the System through Objectives.

"The sole purpose of a business is to maximize profits for shareholders." – Milton Friedman

VISION & SCOPING
BUSINESS ANALYSIS
DATA ENGINEERING
MODEL DEVELOPMENT
DEPLOYMENT
MONITORING


There is vested interest in the AI hype. Expectation Management is crucial.
VISION & SCOPING
BUSINESS ANALYSIS
Starting next AI adoption is hard work and an investment.
DATA ENGINEERING
MONITORING
MODEL DEVELOPMENT
DEPLOYMENT
Returns on investment heavily depend on maturity of adoption.
Development cycles and skills don't develop over-time.

2

Requirements for ML Systems

- **RELIABILITY**
  Correct function at desired level of performance in the face of adversity.
- **SCALABILITY**
  With respect to complexity, throughput, and other types of growth.
- **MAINTAINABILITY**
  Structure workload, infrastructure and tool landscape so that you can keep up.
- **ADAPTABILITY**
  Capacity to discuss potentials for performance improvement and allowing updates without service interruptions.

3

Iterative Process

Developing ML systems is a cycle with a lot of loops and forth!
#ITERATIVE.
VISION & SCOPING
BUSINESS ANALYSIS
DATA ENGINEERING
MODEL DEVELOPMENT
DEPLOYMENT
MONITORING

Objectives
Key Results
Constraints
VISION & SCOPING

Stakeholders: users, managers, humanity

We learn from data, so this is where we start with our craftmanship.
Different services, modalities, formats. - for examples
DATA ENGINEERING

It is all about the labels

Training Data
X
This requires the most ML specific knowledge.
ML
MODEL DEVELOPMENT

Training Data
X
This requires the most ML specific knowledge.
ML
MODEL
DEVELOPMENT
• Feature Engineering
• Model Selection
• Training
• Evaluation
ŷ

You will never reach a point where your system is done.
At some point you need to expose your system.
DEPLOYMENT

Monitoring: stay adaptive

Business analysis closes the loop

4

Framing ML Problems

FRAMING
"The art of simplifying problems before solving them."
- cite me.
Not everything is an ML problem.
But a lot can be framed as one.
VISION & SCOPING
DATA ENGINEERING
MODEL DEVELOPMENT

Classification versus regression

CLASSIFICATION
REGRESSION
0.2
0.8
0.3
class 1
class 2
Threshold to class
Great to enforced classes in X.
Usually you want to reframe in this direction.
binary
multiclass
#Cardinality
Reframe in this direction if possible.

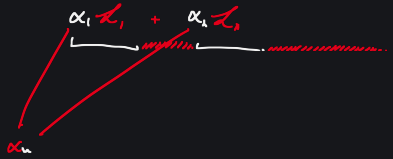
- Restrain from „buying" this with an increasing number of models.
  Hierarchical & Sequential systems are sophisticated but hard to maintain.

→ Talk is cheap. Go and try those things on MNIST to sharpen your own intuition.

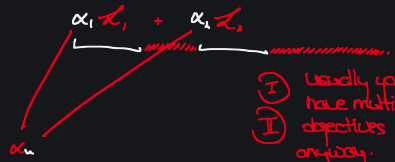
Talk is cheap, sharpen your intuition on MNIST

The objective function defines what optimal is

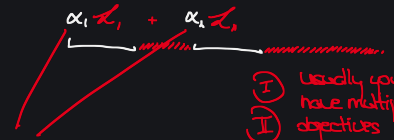
THE EFFECTIVENESS OF DECOUPLING OBJECTIVES.



THE EFFECTIVENESS OF DECOUPLING OBJECTIVES.



THE EFFECTIVENESS OF DECOUPLING OBJECTIVES.



I) Usually you have multiple objectives anyway.

II) This is a clean way to state them explicitly.

Pareto Optimization

Trivial to weight objectives while training.

Everytime you touch is a retrain.

Beautiful opportunity to train two models and adaptively weight during inference.

! Objective Functions require gradients.
For everything else we still need metrics.

5

Mind Versus Data

As data flows through models. During the inference its key
Intelligent architectural designs frozen within the current paradigm. Data-centric.
MODEL success heavily depends on data.
Biggest leverage in managing and improving data Quality and Quantity.
DATA utilization again is tied to compute.

"We don't have better algorithms. We just have better data." - Norvig (Director Search @Google)

Remember how more data is a regularization method, driving generalization of your models?

LECTURE III

Data Engineering Fundamentals

This lecture in one slide

III Data Engineering Fundamentals

Data Sources

Data Formats

Data Models

Data Storage Engines and Processing

Modes of Dataflow

DATA
ENGINEERING
FUNDAMENTALS

Knowing how to handle a growing amount of data

Data Sources
DATA ENGINEERING FUNDAMENTALS
... or knowing how to collect, process, store, retrieve and process an increasing amount of data.

Data Sources
Data Formats, Structures, & Models
DATA ENGINEERING FUNDAMENTALS
...or knowing how to collect, process, store, retrieve and process an increasing amount of data.

Data Sources
Data Formats, Structures, & Models
DATA ENGINEERING FUNDAMENTALS
Data Storage, Retrieval & Processing.
... or knowing how to collect, process, store, retrieve and process an increasing amount of data.

Further reading: Kleppmann 2017

1

Data Sources

Characteristics, purposes, processing methods

DATA SOURCES

- Characteristics
- Purposes
- Processing Methods

... knowing your data lets you use your data more efficiently.

Common practice to log everything you can.
... and you can store.
... and are allowed to.

Log everything you can, store, are allowed to

Common practice to log everything you can.

... and you can store.

... and are allowed to.

Uses Input Data

Halfformatted, any format, requires fast processing.

Common practice to log everything you can.
... and you can store.
... and are allowed to.

User Input Data
Halfformatted, any format, requires fast processing.

System Generated Data
Logs, System outputs, model predictions.

Common practice to log everything you can.

... and you can store.

... and are allowed to.

User Input Data

Halfformatted, any format, requires fast processing.

System Generated Data

Logs, System outputs, model predictions.

Internal Databases

Enterprise applications and assets.

Common practice to log everything you can.
... and you can store.
... and are allowed to.

User Input Data
Halfformatted, any format, requires fast processing.

System Generated Data
Logs, System outputs, model predictions.

Internal Databases
Enterprise applications and assets.

Third Party Data
Data collected on the public. As opposed to customers (1st). Apps, Websites, "Public Spaces", "the Internet".

Third Party Data

"Data Collected on the public. As opposed to customers (1st). Apps, Websites, "Public Spaces, "the Internet".

€

Usually sold after being cleaned and processed.

Third Party Data

"Data Collected on the public. As opposed to customers (1st). Apps, Websites, "Public Spaces, "the Internet".

€

Usually sold after being cleaned and processed.

Data privacy vs. Fingerprinting.

#Apple vs. China.

2

Data Formats

DATA FORMATS

to pick a format, arouses the question on how the data will be used in future.

To pick a format, ask how the data will be used

DATA FORMATS.

to pick a format, answer the question on how the data will be used in future.

Then match the format based on the access pattern, meaning how a system will read or write your data.

DATA FORMATS.

to pick a format, answer the question on how the data will be used in future.

Then match the format based on the access pattern. meaning how a system will read or write your data.

JSON
CSV
PARQUET
Avro
Protobuf
Pickle
!

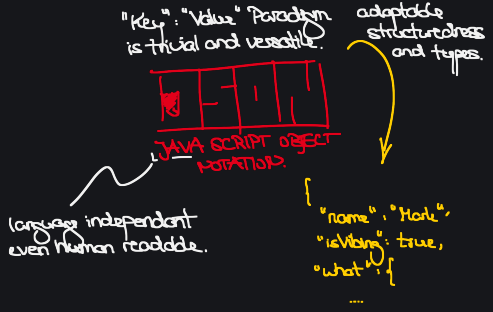
DATA FORMATS.

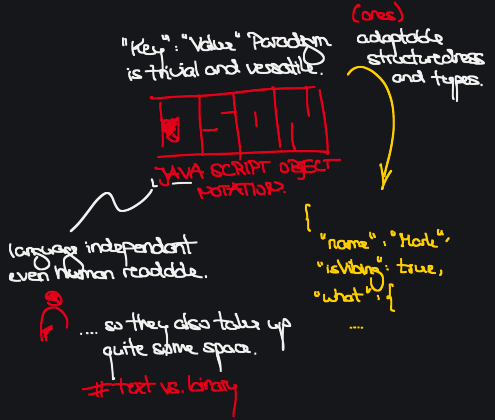
to pick a format, answer the question on how the data will be used in future.

Then match the format based on the access pattern. Meaning how a system will read or write your data.

"Key": "Value" Paradigm
is trivial and versatile.
JSON
JAVA SCRIPT OBJECT
NOTATION

"Key": "Value" Paradigm is trivial and versatile.
(ones)
adaptable structuredness and types.
JAVA SCRIPT OBJECT NOTATION.
{
"name": "Mark",
"isWaving": true,
"what": {
....


"Key" : "Value" Paradigm is trivial and versatile.
(ones) adaptable structuredness and types.
JAVA SCRIPT OBJECT NOTATION.
language independant even human readable.
{
"name" : "Mark",
"isVibing": true,
"what": {
....

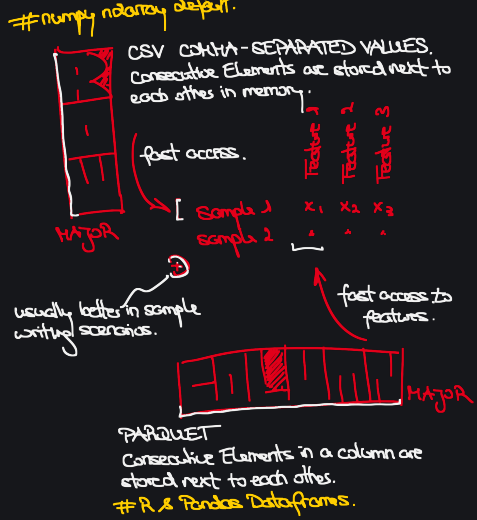

"Key": "Value" Paradigm is trivial and versatile.
(ones) adaptable structuredness and types.
JAVA SCRIPT OBJECT NOTATION.
language independant even human readable.
{
"name": "Mark",
"isWorking": true,
"what": {
....
.... so they also take up quite some space.
#text vs. binary

CSV: consecutive elements side by side in memory

Row major: fast access to samples

Usually better in sample writing scenarios

#numpy ndarray default.
CSV COMMA-SEPARATED VALUES.
Consecutive Elements are stored next to each other in memory.
fast access.
Feature 1
Feature 2
Feature 3
Sample 1
Sample 2
MAJOR
usually better in sample wise processing scenarios.
fast access to features.
MAJOR
PARQUET
Consecutive Elements in a column are stored next to each other.


#numpy ndarray default.
CSV COMMA-SEPARATED VALUES.
Consecutive Elements are stored next to each other in memory.
fast access.
Feature 1
Feature 2
Feature 3
Sample 1
Sample 2
MAJOR
usually better in sample writing scenarios.
fast access to features.
MAJOR
PARQUET
Consecutive Elements in a column are stored next to each other.
#R & Pandas Dataframes.

Text files: assume each character is one byte

Binary files need knowledge to interpret

Assume,
Each character
is one byte.

Text Files

not even one byte but two
characters .. depending on
type.
1000011...

Binary Files
Programs require knowledge
to interpret those.

2x faster to load
6x more compact

Binary: 2x faster to load, 6x more compact

3

Data Models

Data models: how data is represented

Data models affect what your system can solve

RELATIONAL.

each relation is a set of tuples, which are not ordered.

„Think table."

#CSV, Parquet.

NoSQL or nonrelational
or not only relational.
RELATIONAL.
each relation is a
set of tuples, which
are not ordered.
Document Model
Same format: ID.
Think, just rows.
Everything in one place.
„Think tables."
CSV, Parquet.
Graph Model
Relationships modeled
explicitly.

QUICK SIDEQUEST ON QUERY LANGUAGE: SQL.

A declarative language, or „you specify the output you want", not the steps that would take you there (imperative).

DATA MODELS are a pain to maintain, extend or migrate. Unstructured data doesn't adhere to a predefined schema.

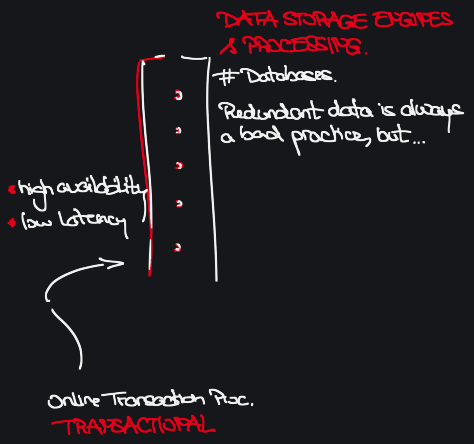
Unstructured data has no predefined schema

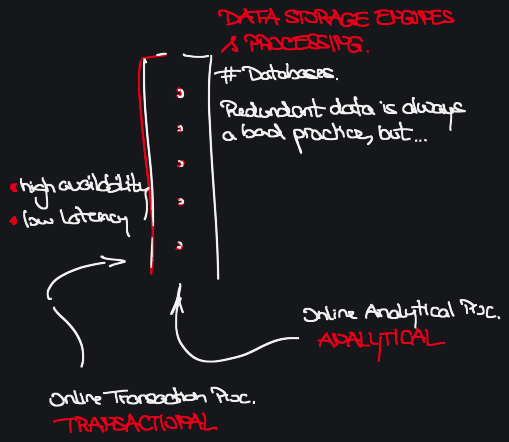
DATA MODELS are a pain to maintain, extend or migrate. unstructured data doesn't adhere to a predefined schema.
But it usually still contains valuable patterns and can be stored effectively #Idea.

4

Data Storage Engines and Processing

Databases: redundant data is a bad practice, but


DATA STORAGE ENGINES & PROCESSING.
#Databases.
Redundant data is always a bad practice, but...
high availability
low latency
Online Transaction Proc.
TRANSACTIONAL


DATA STORAGE ENGINES & PROCESSING.
#Databases.
Redundant data is always a bad practice, but...
high availability
low latency
Online Analytical Proc.
ANALYTICAL
Online Transaction Proc.
TRANSACTIONAL

Online Transaction Proc.

TRANSACTIONAL

Atomicity.

Transaction fails with a single failed step.

Online Transaction Proc.

TRANSACTIONAL

- Atomicity.
  Transaction fails with a single failed step.
- Consistency.
  All transactions follow the same rules

Online Transaction Proc.

TRANSACTIONAL

Atomicity.
Transaction fails with a single failed step.

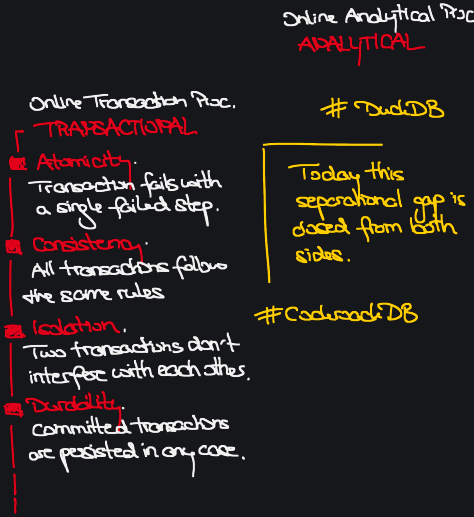
Consistency.
All transactions follow the same rules

Isolation.
Two transactions don't interfere with each other.

Online Transaction Proc.

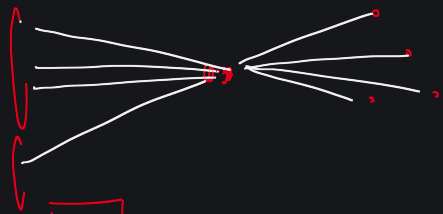
TRANSACTIONAL

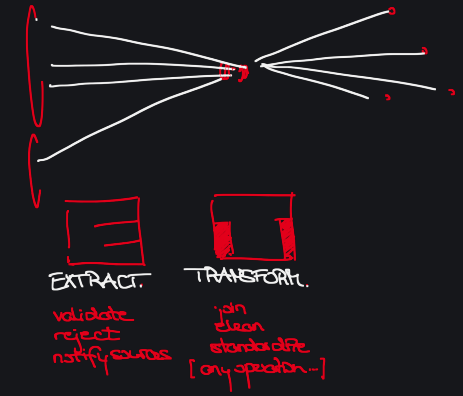
1. Atomicity.
   Transaction fails with a single failed step.
2. Consistency.
   All transactions follow the same rules
3. Isolation.
   Two transactions don't interfere with each other.
4. Durability.
   Committed transactions are persisted in any case.


Online Transaction Proc.
TRANSACTIONAL
Atomicity.
Transaction fails with a single failed step.
Consistency.
All transactions follow the same rules
Isolation.
Two transactions don't interfere with each other.
Durability.
Committed transactions are persisted in any case.
Online Analytical Proc.
ANALYTICAL
#DuckDB
Today this separational gap is closed from both sides.
#CockroachDB

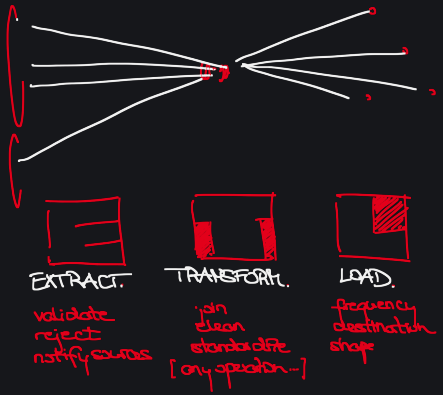
Paradigm to decouple storage from compute,
to reach general purpose data.

Paradigm to decouple storage from compute, to reach general purpose data.

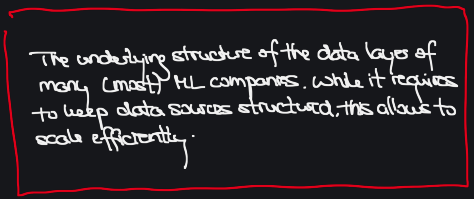



Paradigm to decouple storage from compute,
to reach general purpose data.
EXTRACT.
validate
reject
notify issues
TRANSFORM.
join
clean
standardize
[any operation...]

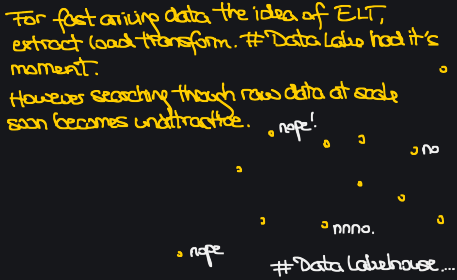
Paradigm to decouple storage from compute, to reach general purpose data.



The structure of most ML data layers



For fast arriving data the idea of ELT, extract load transform. #Data lake had it's moment.
However searching through raw data at scale soon becomes unattractive.


For fast arriving data, the idea of ELT, extract load transform. #Data Lake had it's moment.
However searching through raw data at scale soon becomes unattractive.
nope!
no
nnnno.
nope
#Data Lakehouse....

5

Modes of Dataflow

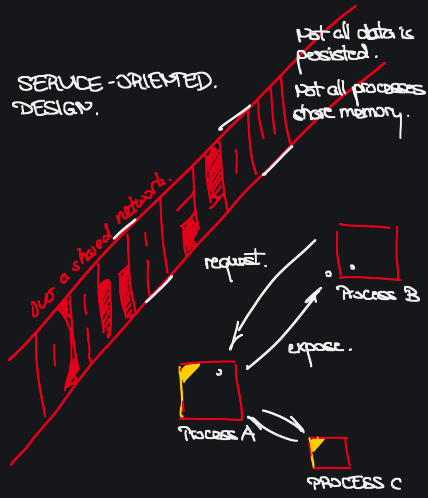
Not all data is persisted, not all memory shared

Write to a database: persisted and shared

Read from the database, slow

SERVICE-ORIENTED DESIGN.
Not all data is persisted.
Not all processes share memory.
Duo a shared network.
Process B
Process A

Request and expose


SERVICE-ORIENTED.
DESIGN.
Not all data is persisted.
Not all processes share memory.
DATAFLOW
Use a shared network.
request.
expose.
Process A
Process B
PROCESS C

Microservice Architecture
as a design principle.
Develop, test, maintain
independently. #API-first. #REST

Microservice Architecture
as a design principle.
Develop, test, maintain
independently. #API first. #REST

FURTHER READING:
Building Microservices by Sam Newman.

Microservice Architecture
as a design principle.
Develop, test, maintain
independently. #API first. #REST

FURTHER READING:
Building Microservices by Sam Newman.

... and then design your org chart around your architecture.

Many processes, many data paths

Passing data gets complicated and stale fast

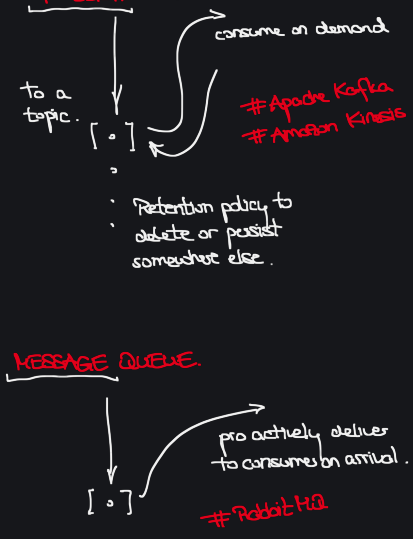
EVENT-DRIVEN.
Not all data is persisted.
Not all processes share memory.
DATAFLOW

Processes publish events to a bus

The broker, usually in memory

Publish subscribe: to a topic, consume on demand

Retention policy. Apache Kafka, Amazon Kinesis


PUBLISH SUBSCRIBE.
consume on demand
to a topic.
#Apache Kafka
#Amazon Kinesis
Retention policy to delete or persist somewhere else.
MESSAGE QUEUE.
pro actively deliver to consumers on arrival.
#RabbitMQ

This is all for streaming data, as opposed to historical, batch processed data.

This is the last slide.
Look how far you've come.

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